

CSc 4710/6710: Database Systems

Spring 2026, Assignment 2

Due date: Tuesday, 04/07/2026

During submission, you must mention your name, Panther ID, and Course number (4710/6710).

Problem 1 [15 = 5*3]

Consider a relation R with attributes ABCDE and the following functional dependencies:

$$F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D, D \rightarrow E\}$$

1. Compute the attribute closure of $\{A\}$.
2. Compute the attribute closure of $\{A, C\}$.
3. Identify all candidate keys of the relation.

Problem 2 [10 = 5*2]

Consider a relation R(P, Q, R, S) and functional dependencies:

$$F = \{PQ \rightarrow R, R \rightarrow S, S \rightarrow Q\}$$

1. (5) What are all the candidate keys for R?
2. (5) Is the relation in BCNF? Justify your answer.

Problem 3 [10 = 5*2]

Consider a relation R(A, B, C, D) with:

$$F = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$$

1. Is the relation in 2NF? Explain.
2. Is the relation in 3NF? If not, decompose it into 3NF.

Problem 4 [20=5*4]

Consider the following tables-

Course (CourseID, Title, Credits)

Enrollment (StudentID, CourseID, Grade)

Student (StudentID, Name, Major)

Write the following queries in SQL:

- a. Write a query to find the names of students who have enrolled in at least 3 different courses.
- b. Write a query to find the names and majors of students who have taken all the courses worth 3 credits.
- c. Write a query to find the student(s) who have the highest average grade
- d. Write a query to find students who have never taken any course with fewer than 3 credits.

Problem 4 [15=5*3]

Consider the following table-

Movie(MovieID, Title, Genre, Rating)

Review(ReviewID, MovieID, ReviewerName, Score)

- a. Find the titles of movies that have a higher average review score than every movie in the 'Horror' genre.
- b. List all movies that have never been reviewed.
- c. Find the reviewer(s) who have given a review to every movie in the database.